

**HIGH LEVELS OF HISTIDINE-RICH GLYCOPROTEIN AND
THROMBOTIC DIATHESIS**

REPORT OF TWO UNRELATED FAMILIES

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Abstract Two new families with history of thrombosis and high levels of histidine-rich glycoprotein (HRG) are described. The proband of family 1 died of massive pulmonary embolism at age 34. Among his relatives, the mother and the maternal grandmother had suffered from deep and superficial vein thrombosis in their youth. A maternal aunt had several episodes of superficial vein thrombosis (SVT). High levels of HRG were found in the mother, three siblings and two nephews. In the second family, the proband suffered from spontaneous deep vein thrombosis (DVT) at age 24. The paternal grandmother and a paternal aunt had several episodes of SVT and DVT. Also in this family, high levels of HRG cosegregated with thrombotic symptoms. These new families confirm that genetically transmitted high levels of HRG could be associated to familial and juvenile thrombophilia.

Histidine-rich glycoprotein (HRG) is a non-enzymatic plasma protein which interacts with the high affinity lysine-binding site of plasminogen (1) and binds heparin with high affinity (2). Through these interactions, HRG reduces the amount of free

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plasminogen that binds to fibrin clot and lessens the amount of heparin available for the enhancement of thrombin-antithrombin III reaction. Furthermore, it has been demonstrated that HRG is present in the α -granules of platelets from where it is released after thrombin stimulation (3). In this way, platelet-released HRG may achieve a high local concentration and play a significant role in modulating fibrinolysis in the microenvironment of the platelet plug.

Conceivably, high levels of this protein could be associated to thrombophilia and, in fact, recently a kindred with familial elevation of HRG and thrombotic symptoms in the propositus has been reported (4). Subsequently, additional cases have been published (5-8), sometime associated with high levels of plasminogen-activator inhibitor (PAI-1) (7,8).

In this study, we report two new families with autosomal dominant transmission of very high levels of HRG and evidence of thrombotic diathesis.

MATERIALS AND METHODS

Coagulation assays

Activated partial thromboplastin time (APTT), prothrombin time (PT), thrombin time (TT) and fibrinogen were assayed on a laser-automated instrument (ACL, Paderno Dugnano, Italy), using Instrumental Laboratory (IL) reagents and according to the manufacturer instructions. Antithrombin III (AT-III) activity was measured as heparin cofactor activity using the chromogenic substrate S-2238 (Kabi, Stockholm). Total and free protein S antigen was assayed by rocket immunoelectrophoresis (Asseraplate, Stago, Paris). Protein C activity was determined using the chromogenic substrate S-2366 (Kabi). Plasminogen and α -2-antiplasmin activities were measured using an amidolytic method with the chromogenic substrate S-2251 (Kabi). Heparin cofactor (HC) II activity was measured after activation with dermatan sulfate using the chromogenic substrate 2 ACOH.H-D-CHG-Gly-Arg-pNA (Channel Diagnostics, supplied by Cabru, Milan). PAI-1 activity was assayed using a chromogenic substrate (Kabi).

HRG antigen was measured by electroimmunoassay (Asseraplate, Stago). Crossed immunoelectrophoresis (CIE) of a mixture of normal plasma or serum HRG with that of subjects with high levels of HRG was carried out using 2.1 % of antiserum (Stago, Paris) in the second dimension. CIE was performed also with heparin (Liquemin, Roche, Basel) (16 U/mL, final concentration) in the first dimension.

CASE REPORT

1) FAMILY B

The relatives (mother, an uncle, the 3 siblings, the 3 nephews and one niece) of a man died at age 34 of massive pulmonary

embolism after deep vein thrombosis (DVT) were investigated for coagulation defects predisposing to thrombophilia (Figure 1). The propositus (III. 4) at age 25 suffered from superficial vein thrombosis (SVT) and at age 34 was admitted to the hospital for spontaneous DVT. Intravenous heparin (25,000 units/day by continuous infusion) was started. After few hours, severe dyspnoea developed and a clinical diagnosis of pulmonary embolism was made. Heparin was augmented (37,500 U/day) and recombinant tissue-type plasminogen activator (100 mg over two hours) was started. Despite this treatment, the patient died for cardiac arrest. Autopsy revealed DVT of right leg and massive pulmonary embolism with upper right lobe infarction. In addition, signs of recent myocardial infarction were documented. Unfortunately, no plasma samples were available for coagulation studies.

The mother (II.3, aged 68) suffered from DVT at age 29 during puerperium after her third pregnancy and from repeated SVT episodes. The maternal grandmother (I.2) had DVT after delivery at age 32 and 35 as well as several episodes of SVT. A maternal aunt (II.2, aged 69) had several episodes of SVT. The maternal uncle (II.1, aged 71), the two sisters (III. 2, aged 38, and III.3, aged 36), the brother (III.1, aged 40) and 3 nephews (IV.1, aged 4; IV.3, aged 14, and IV.4, aged 12) and the niece (IV.2, aged 6) were all asymptomatic.

2) FAMILY M

The propoita (III.1, Figure 2) was referred for spontaneous DVT confirmed by impedance pletysmography at age 24. She had had no surgical intervention or pregnancy. A paternal aunt (II.3, aged 56) had suffered from DVT after leg fracture, confirmed by venography, and the grandmother (I.2) from DVT after delivery and from repeated SVT. The two sisters, aged 23 and 21, and the father, aged 50, were asymptomatic.

RESULTS

Table I summarizes the results of coagulation and fibrinolytic assays of the family members tested. PTT, PT and TT were in the normal range (not shown). As can be observed, the screening failed to detect abnormalities of the known proteins whose deficiency or abnormality is commonly associated with familial or juvenile thrombophilia.

Very high levels of HRG antigen were consistently demonstrated over a 3-year period in several members of the two families (Table II). In both the families, thrombotic symptoms were exclusively present in members with high levels of HRG. HRG level was also established in 20 children aged 10-15 and found to be significantly lower than adults (mean $83 \pm 20\%$ vs $103.5 \pm 18\%$; $P < 0.001$). This allowed the inclusion among the affected members also of IV.3, aged 14, and IV.4, aged 12, in family B. Crossed immunoelectrophoresis of HRG in plasma, with and without heparin, showed a normal migration pattern (data not shown). Additional

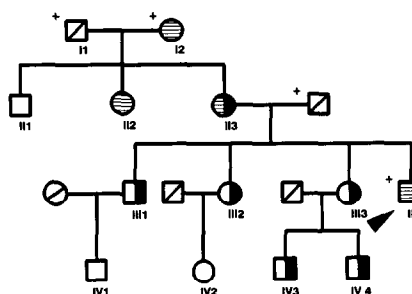


FIG. 1.

Pedigree of family B.

- ◻/◻ asymptomatic, not tested
- ◼/◼ history of thrombosis, not tested
- ◻/◻ asymptomatic, normal HRG
- ◼/◼ asymptomatic, high levels of HRG
- ◼/◼ history of thrombosis, high levels of HRG
- ▼ propositus
- + deceased

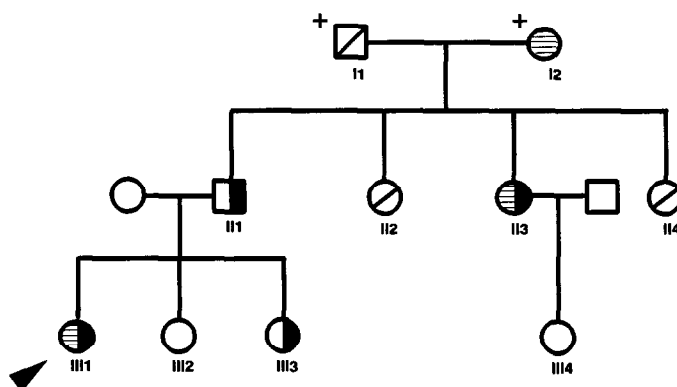


FIG. 2.

Pedigree of family M (for symbols, see Fig. 1).

CIE studies carried out after diluting propositi plasma with buffered saline (HRG level measured in the mixture was about 110%) or with equal amount of normal plasma also failed to show any abnormality (data not shown).

TABLE I

**Laboratory parameters in the members
with high levels of histidine-rich glycoprotein**

TEST	SUBJECTS				
	<u>FAMILY B</u>				
	II.3	III.1	III.2	III.3	Normal range
Fibrinogen (mg/dL)	320	240	200	180	150 - 400
Antithrombin III (%)	128	103	112	102	75 - 140
Protein S total (%)	98	87	80	83	75 - 145
Protein S free (%)	89	117	95	108	69 - 136
Protein C (%)	131	103	96	106	73 - 141
Heparin Cofactor II (%)	162	83	119	77	70 - 129
Plasminogen (%)	129	95	119	104	72 - 148
α -2-antiplasmin (%)	131	114	115	113	62 - 154
PAI-1 (AU/mL)	11	5	4	9	1 - 13
	<u>FAMILY M</u>				
	II.1	II.3	III.1	III.3	Normal range
Fibrinogen (mg/dL)	180	240	180	220	150 - 400
Antithrombin III (%)	108	104	103	109	75 - 140
Protein S total (%)	112	155	89	95	75 - 145
Protein S free (%)	100	85	91	123	69 - 136
Protein C (%)	99	112	90	103	73 - 141
Heparin Cofactor II (%)	94	90	97	100	70 - 129
Plasminogen (%)	89	100	86	89	72 - 148
α -2-antiplasmin (%)	100	103	96	103	62 - 154
PAI-1 (AU/mL)	8	12	6	9	1 - 13

DISCUSSION

Very high levels of plasma HRG were found in six relatives of a patient died at age 34 for massive pulmonary embolism after DVT (Family B, Figure 1) and in the proposita and three relatives of family M (Figure 2). The mother, a maternal aunt and the maternal grandmother of the propositus of family B had suffered from juvenile episode of DVT and several episodes of SVT. The mother also had very high levels of HRG. The two sisters, the brother and two nephews of the propositus showed the same laboratory abnormality without evidence of thrombotic symptoms. In family M, high levels of HRG were present in the proposita and a paternal

TABLE II

Levels of histidine-rich glycoprotein in the family members tested

	HRG (%)	Thrombosis
<u>Family B</u>		
II.3	190	YES
III.1	250	NO
III.2	188	NO
III.3	210	NO
*IV.3	155	NO
*IV.4	160	NO
II.1	98	NO
IV.1	100	NO
IV.2	66	NO
<u>Family M</u>		
II.1	200	NO
II.3	190	YES
III.1	260	YES
III.3	250	NO
III.2	76	NO
III.4	95	NO
Normal adults		
(n=40)	103	
range	68 - 136	
*Children (aged 10-15)		
(n=20)	83	
range	53 - 113	

 The results are the average of 3 to 5 determinations

aunt, with thrombotic symptoms, and in the father and a sister of the proposita. In all the subjects carrying the laboratory abnormality, elevated levels of HRG were stable over a 3-year period of study. An autosomal dominant pattern of inheritance of the laboratory abnormality is thus evident. Moreover, thrombosis was evident only in patients with high levels of HRG.

Interestingly, the propositus reported by Engesser et al (4) suffered from myocardial infarction, at age 53, albeit he was still on warfarin therapy for a previous episode of pulmonary embolism. In the propositus of family B, at autopsy, it was possible to ascertain a recent postero-septal myocardial infarction. No other risk factors for arterial thrombosis (dyslipidemia, diabetes, smoke, hypertension) were evident. An additional patient suffering from angina pectoris and with

familial elevation of HRG has been recently described (6). Thus, the association of severe thromboembolic events with arterial thrombosis could be a peculiar feature of this abnormality; however, other families should be reported to assess the true significance of this finding.

Falkon et al. (7) reported a family with high levels of HRG with thromboembolism present only in the proposita, in whom high levels of PAI-1 activity were also evident. An additional family has been recently reported in abstract form (8), with thrombosis in members with elevated levels of HRG and PAI-1. In our patients, PAI-1 activity was in the normal range.

HRG levels are higher in adults than in children (9) and thus it is important to establish the normal ranges according to age. In fact, in our families it was possible to include two nephews of the propositus of family B by adopting a separate range for children.

Apart from mild liver cirrhosis (10), acquired high levels of HRG have not been documented in clinical situations, whereas, on the contrary, low levels have been demonstrated in several conditions, especially sepsis or liver failure (11-13). None of our subjects had clinical or laboratory evidence of liver impairment. HRG is able to lower the amount of free plasminogen and its reduction, as demonstrated by Jespersen et al. (14) in women taking oral contraceptives with low estrogen content, induces an increase of free plasminogen thus enabling an increased binding rate of plasminogen to fibrin. Very high levels of HRG thus could complex large amounts of plasminogen and critically depress fibrinolysis, also considering its release by platelets, following thrombin stimulation, in sites where fibrin formation is ongoing (3).

The reason for the presence in the same family of asymptomatic members who experienced situations at risk, such as pregnancy and surgical interventions, without developing symptoms and affected subjects with severe thrombotic symptoms is not clear but this pattern (low penetrance) has been reported for other families with inherited hemostatic abnormalities predisposing to thrombosis (15).

Following the demonstration of the existence of families with genetically transmitted high levels of HRG, prophylaxis of rethrombosis and prevention in situation at risk should be considered.

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